

What do we mean by *language*?

A pluralist map for the language sciences

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Abstract

The word *language* names several distinct objects across the language sciences: capacity, idiolect, communicative practice, population convention, named sociohistorical object, and formal system. Pluralist accounts (especially Santana’s distinction among psychological, social, and abstract ontologies) show why monism fails but don’t say when a proposed sense licenses inference or when evidence about one sense can travel to another. This paper adds a homeostatic property cluster (HPC) discipline: a sense earns realist standing where its properties are stabilized by identifiable mechanisms and project for a field’s purposes; where the mechanisms diverge, the shared label doesn’t license transfer. The map distinguishes three categories proper (population convention, named sociohistorical object, abstract formal system) from three neighbourhoods of substantively coupled constituent categories (capacity, idiolect, communicative practice). The payoff is diagnostic. The Casten / Pinker exchange over genomic evidence for spoken-language ability shows how developmental verbal-performance evidence gets read as evidence about a uniquely human language faculty; similar slippages arise when census recognition is read as convention, formal models as cognitive architecture, or analyst diagnostics as the object. The umbrella label remains useful for coordination, but the load-bearing scientific units are the categories and neighbourhoods, not the umbrella.

I A SLIPPAGE AND A QUESTION

In April 2026, Casten et al. (2026) reported that a small fraction of the human genome carries an out-sized share of variation in spoken language ability. Their target measure, the FI factor, is anchored by sentence repetition and aggregates morphosyntactic, phonological, working-memory, lexical, and motor contributions to verbal performance. In a community-based cohort of 350 children assessed longitudinally from kindergarten through fourth grade, a polygenic score restricted to human-ancestor quickly-evolved regions (HAQERs, less than 0.1% of the genome) showed an effect of $\beta = 0.18$ on FI, no effect on nonverbal IQ, and replication across more than 100,000 participants in three further cohorts. Cross-species analysis of 170 nonprimate mammalian genomes found that HAQER-like sequence similarity correlates with vocal-learning ability across 49 vocal-learning species and 121 non-vocal learners.

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Three days later, on 25 April 2026, Steven Pinker cited the paper on X (Pinker, 2026) as vindication of Pinker and Bloom (1990). The Pinker–Bloom thesis, as Pinker summarizes it on the day, is that the human language faculty is “a genetically complex trait, distinct from general cognition, which evolved gradually by natural selection of many regulatory genes over a long period.” The Casten paper’s polygenic-score, ancient-DNA, and selection-signal results, Pinker writes, “suggest we were right.” The reading was widely circulated and approvingly relayed. This is good evidence.

It isn’t evidence for the larger claim.

What Casten et al. measured is a developmental verbal-performance phenotype anchored by sentence repetition: a score that loads on attention, working memory, and lexical access alongside grammatical processing. The HAQER-like regulatory regions are shared across nonprimate mammalian vocal learners (cetaceans, pinnipeds, bats); whether they carry human-specific tuning is a separable question their evidence doesn’t address. Pinker–Bloom’s faculty is supposed to be *distinct from general cognition* and human-specific. The update the evidence licenses runs to the shared-mammalian regulatory backbone of the broad faculty, not to the distinctively human complex Pinker–Bloom were arguing for; the shared label *language* obscures the gap.

The pattern recurs. The word *language* does work for several recurrent senses across linguistics, cognitive science, anthropology, philosophy, semiotics, law, and education. Here a faculty, there a convention; here a phenotype, there a natural object; here a set of strings, there a situated practice. Each is real, well-studied, and distinct in its projection target; none licenses inferences about the others without further argument. The Casten / Pinker case shows the recurrent pattern in miniature.

The question, then, is how we should use the word *language* across fields, given that it does work for senses that don’t all license the same inferences. Two answers fall short. Monism picks one sense and treats the rest as misapplications: Chomsky (1986, p. 26) dismisses social senses as “artificial, somewhat arbitrary, and perhaps not very interesting constructs”, but Santana (2016) shows the positive arguments for each sense succeed and the negative arguments for ruling out the others generally fail. Unconstrained pluralism accepts the senses as a list and leaves it. Santana rules that out too: the senses interact, and confirmation, idealization, and theory choice turn on which sense a researcher is committed to.

This paper takes a third route. The senses cluster: they are recurrent, recognizable, and rough enough at the edges that working scientists already treat them as different. Each earns realist standing *for the field purposes its mechanisms underwrite*; when a claim asks one to project beyond that range, the inference needs an additional bridge. Boyd’s homeostatic-property-cluster (HPC) framework supplies the discipline (Boyd, 1991, 1999). The contribution is the synthesis: a more granular taxonomy than Santana’s three ontologies, plus the diagnostic that lets us say when claims about one element smuggle in claims about another. Section 9 works through the Casten / Pinker case in detail.

The map proposed here foregrounds six elements: three neighbourhoods (capacity, idiolect, communicative practice) and three categories (population convention, named sociohistorical object, abstract formal system). The criterion for distinguishing the cases, set out in §2, runs through shared stabilizers, projection-target overlap, and perturbation coupling. The taxonomy is open: new categories or neighbourhoods are admissible if they pass the projectibility and homeostasis tests. Section 10 returns to the umbrella label *language* as a coordinative term across fields. The elements under it are what working science actually projects from.

2 PLURALISM, AND WHAT HPC ADDS

The closest extant precedent for what follows is Santana (2016). Santana inventories the candidate ontologies that linguists work with – psychological (Chomskyan i-language; psycholinguistic processing), social (Lewis-style convention; Labov’s speech communities; Devitt’s externalism), and abstract (Katz’s Platonism; Montague semantics; formal-language theory) – and shows that the positive arguments for each succeed while the negative arguments for ruling out the others generally fail. The result is a pluralist ontology of language: more than one of these classes is real, and which one a researcher’s claims target depends on what kind of inferences they’re trying to license.

Santana frames this as three criteria a candidate ontology has to meet (existing in a form accessible to scientific study, being a primary object of study for some subfield, and being reasonably called *language*) plus two strictures: no isolationism (research on each ontology bears on the others, and their relations are themselves worth investigating) and no agnosticism (in any research context, the scientist has to be explicit about which ontology they’re committed to, because confirmation, idealization, and theory choice turn on the commitment).

Santana’s framework gets the lay of the land right. What it doesn’t supply is a discipline for testing whether a given category earns its realist standing in the work it’s asked to do. Three things aren’t yet in hand: a test for projectibility (does the category license inferences from observed to unobserved members?), a test for homeostasis (do identifiable mechanisms sustain it, with perturbation predicting patterned decay?), and a way of diagnosing slippage (is evidence about one category being read as evidence about another, with no bridging argument, only because both are called *language*?).

Boyd (1991, 1999) supplies the missing piece. The HPC framework was developed for natural kinds whose members share a profile of co-occurring properties without a defining essence. A CLUSTER is a profile of properties that tend to go together. A cluster earns CATEGORY status, in the realist sense needed here, when that profile projects: learning about observed cases supports counterfactual expectations about unobserved ones. Three checks discipline the claim. The cluster check asks whether property covariance holds. The homeostasis check asks whether identifiable stabilizers explain that covariance and predict patterned change when perturbed. The projectibility check asks whether the cluster supports the inferences a field wants to draw (see also Khalidi, 2013). Passing the checks for one field’s purposes doesn’t entail passing them for another, and FIELD-RELATIVE PROJECTIBILITY lets two fields carve overlapping extensions differently; the *Cantonese* case in §9 works the asymmetry concretely. Together the three checks tell us what projects, what holds together, and what has slipped.

The framework applies recursively. At any level, a cluster either projects as one category or it doesn’t, with projection then living at finer grain among the constituent categories. Three diagnostics distinguish single-category status from a coordinative label: shared stabilizer set, projection-target subsumption, and perturbation coupling. Perturbation evidence comes in three forms (counterfactual experiment, historical natural experiment, and comparative-typological evidence), with weight scaling to the rigour of the demonstration; intuition about coupling, without any of the three, doesn’t satisfy the test. Perturbation coupling itself admits of degree: symmetric coupling licenses full single-category confidence, while asymmetric coupling preserves single-category status at lower confidence and narrower projection (worked in §8). When the sub-divisions answer no on all three diagnostics, the named label is coordinative rather than a single category. Coordinative labels come in two degrees of coupling. A NEIGHBOURHOOD has substantively coupled constituent categories: shared scale of inquiry, shared explanandum, and mechanistic or developmental relations among them. Capacity’s

constituent categories share individual-internal cognitive and biological architecture; idiolect's share the individual-scale explanation of what speakers know; communicative practice's share situated joint-action as their object.

The verdicts have to be earned, not asserted. Table 1 runs the diagnostic on each of the six map elements; §§3–8 work the cases case by case. The asymmetry that demotes capacity, idiolect, and communicative practice isn't about which stabilizer sets count as "unified enough"; it's about whether the strands operate at scales where perturbation actually couples them.

Table 1: Test verdicts for the six map elements.

Element	Stabilizers	Perturbation coupling	Verdict
Population convention	Alignment, transmission, metapragmatic feedback	Coupling is strongest where mutual expectation, network density, and indexical uptake condition the same population regularity (Eckert, 2008; Lewis, 1969; Milroy, 1987)	Category
Named sociohistorical	Institutional codification, count-noun ideology, looping feedback	Case-specific: institutional ratification, identity uptake, and historical-lineage claims shift together only where a named object is actively maintained; §9 works the Cantonese case	Category
Abstract formal	Mathematical inheritance, definitional stipulation, disciplinary pipelines	Asymmetric: formal-language theory and semantics propagate; NLP engineering doesn't always perturb back	Category (borderline)
Capacity	FLN architecture, broad faculty, developmental phenotype, vocal-learning, impairment	Decoupled across scales: a developmental finding doesn't predict an architectural conclusion	Neighbourhood
Idiolect	I-language (UG + maturation), Devittian command, individual variation	Different ontologies; perturbing exposure doesn't perturb architectural commitments	Neighbourhood
Communicative practice	Speech-act rules, interactional alignment, appropriateness, metapragmatic typification	Enregisterment can fragment without felicity conditions shifting; scales decoupled	Neighbourhood

The verdicts come with confidence calibrated to the rigour of the perturbation evidence: high for population convention at the conditioned-regularity grain, moderate and case-specific for named sociohistorical, asymmetric for abstract formal (and explicitly so – §8 works the asymmetry). A verdict isn't a binary stamp but a current-best assessment whose confidence the perturbation evidence sets.

Substantive coupling can include partial co-constitution: one constituent category may help build, constrain, or reveal another. That still doesn't make the neighbourhood a single category unless the coupled parts share a stabilizer set, perturb together, and project to a subsuming target. An UM-BRELLA has loosely coupled elements: shared field of inquiry and research community, but different

scales, different mechanism types, no shared explanandum. The umbrella label *language*, treated in §10, organizes capacity-research alongside formal-language theory and population-convention work, across scales and mechanism types that don't share a target. Khalidi's permissive line would call neighbourhoods themselves cluster categories at coarser grain; this paper adopts the stricter Boyd-style line, where coordinative labels organize categories without themselves projecting as one.

The verdicts are revisable in both directions. A neighbourhood can be promoted if a unifying stabilizer set is later demonstrated (Maxwell unifying electrostatic and magnetic phenomena, the modern synthesis unifying Mendelian inheritance and Darwinian selection); a category can be split if its sub-divisions turn out not to share the assumed mechanism. On present evidence, capacity (§3), idiolect (§4), and communicative practice (§5) come out as neighbourhoods; population convention (§6), named sociohistorical objects (§7), and abstract formal systems (§8) satisfy single-category status.

The contribution of this paper is the synthesis. The taxonomy below is finer-grained than Santana's three classes: each of the social, psychological, and abstract classes contains more than one category, with some passing single-category status and others resolving into a neighbourhood of related constituent categories with distinct projection targets. SLIPPAGE across categories becomes diagnosable in HPC terms as a mismatch between the projection target a researcher's evidence actually licenses and the target their conclusion is reaching for.

A short example shows the subtraction at the level of inference. Santana's three criteria license treating *English subject-verb agreement* as a real object of inquiry, and the two strictures hold: claims about it bear on neighbouring features (no isolationism), and researchers studying it commit to the object (no agnosticism). Santana's framework licenses claims of the form "English speakers display existential-*there* agreement at rate R". HPC's homeostasis check rules out some such claims. Section 6's audit shows the move: the pooled rate conceals two register-conditioned subsystems whose stabilizers differ, and the covariance under the pooled label fails to reconstitute under perturbation. Santana's framework permits the inference; HPC's discipline tells you which grain it actually projects from. Sections 6–9 catalogue cases like this across the map.

A related failure occurs when the operational proxy replaces the target. Each element gets operationalized differently in research practice: corpora and treebanks for the social elements, competence and performance idealizations for capacity, formal-language-theoretic objects for the abstract category, ethnographic and conversation-analytic transcription for communicative practice. Claims of the form "the corpus IS the language" or "competence IS the language" substitute the analyst's diagnostic for the element it's supposed to track. Section 9 catalogues these alongside the across-the-map cases.

3 CAPACITY

CAPACITY is what the literature gathers about what humans can do with language: acquire, process, perceive, produce, and store linguistic material. Pinker and Bloom argue it's a complex adaptation (Pinker & Bloom, 1990); Hauser et al. (2002) partition it into broad and narrow faculties; developmental-genetic and clinical-genetic studies aim at it when they look for individual differences. The constituent categories foregrounded here recur in the literature with stabilizing mechanisms operating at different grains, licensing different inferences, and not sharing a single homeostasis. Capacity is best treated as a neighbourhood of related constituent categories at the individual-internal scale.

One constituent category is the abstract computational architecture (generative biolinguistics): what is uniquely human and uniquely linguistic in the system that maps form onto meaning. Hauser et al. (2002) call this the faculty of language in the narrow sense (FLN) and propose that it consists of recursive computation plus the interfaces with sensory-motor and conceptual-intentional systems. Pinker and Jackendoff (2005) contest this characterization, arguing that phonology has discrete infinity but not technical recursion, that words encode language-specific syntactic information not captured by recursion alone, and that case, agreement, and most of the syntactic apparatus beyond hierarchical phrase grouping are language-specific. Stabilizers proposed for this constituent category run from genetic endowment, often framed as Universal Grammar (UG), through cognitive-architectural primitives like Merge to the convergent product of multiple selected components. Observed in this constituent category, *P* (a structural feature like binding or displacement) licenses expected *Q* (a typological universal or an acquisition-order prediction) in unobserved cases; defeated if *D* (the feature is acquired by general-cognitive mechanisms, varies cross-linguistically in ways UG-style theories can't accommodate, or appears outside language entirely).

A second constituent category is the broad faculty (comparative cognitive biology): the sensory-motor system, the conceptual-intentional system, and their interfaces with the architectural constituent category above. Hauser et al.'s claim is that most of the broad faculty is shared with other species, including capacities for vocal imitation, categorical perception of speech-like sounds, theory of mind, and conceptual representation of objects, agents, and events. The methodological backbone here is comparative biology. A capacity in species *X* projects to the same capacity in species *Y* via the comparative-method warrant: homology under phylogenetic continuity predicts inherited recurrence; convergent evolution under similar selection predicts analogous recurrence. Defeated if the capacity proves uniquely human, fails under denser sampling, or tracks a phylogenetic confound.

A third constituent category is the developmental-acquisition phenotype (developmental psycholinguistics, clinical linguistics). Children acquire languages at species-specific rates and modes, regardless of modality (spoken or signed), with sensitive-period effects on phonology, vocabulary, and morphosyntax. The Casten et al. *F1* measure introduced in §1 sits squarely here: a longitudinal aggregate of sentence repetition, working memory, attention, and lexical retrieval scores. Stabilizers are developmental input, neural and cognitive maturation, and modality-general abstraction. An acquisition pattern in one cohort projects to similar outcomes in unobserved children with comparable input, or to replicable associations with a genetic variant; defeated if replication fails or the measure tracks general cognitive variance rather than language-specific trajectory.

A fourth constituent category is vocal-learning regulatory architecture (cross-species comparative biology). Convergent regulatory specializations for vocal learning show up in genomic comparisons that control for phylogenetic relatedness. Within placental mammals, Wirthlin et al. (2024) report convergent regulatory specializations across the four mammalian vocal-learning lineages (bats, cetaceans, pinnipeds, and humans), drawing on 215 mammalian genomes. The HAQER-like sequences from Casten et al. (2026), correlated with vocal-learning ability at the cross-taxon scale, sit alongside. Stabilizers are convergent regulatory tuning, transcription-factor binding affinities, and developmental constraints on vocal-learning-related regulatory architecture. A regulatory architecture in species *X* projects to a similar architecture in species *Y* conditional on convergent selection; defeated if the convergence claim fails under denser sampling or tracks a phylogenetic confound.

A fifth constituent category is impairment phenotype (clinical linguistics, language genetics). Spe-

cific Language Impairment, FOXP2-related disorders, dyslexia, and developmental stuttering pick out variation that is heritable and partly localizable to specific genetic loci, but the impairments span multiple grammatical and processing components rather than knocking out a single circuit. The argument from impairment phenotype is one of Pinker and Jackendoff's pieces of evidence against the recursion-only hypothesis: heritable language impairments aren't heritable recursion impairments. Stabilizers here are specific genetic loci with measurable effects, polygenic background, and developmental cascades from molecular variation to behavioural phenotype. A specific genetic variant projects to a related deficit in unobserved cases sharing the variant; defeated if the phenotype is mechanistically heterogeneous or the effect doesn't replicate.

The constituent categories partly co-constitute one another: developmental acquisition tunes the architecture it later reveals, broad-faculty systems constrain what the architecture can be, comparative-biology mechanisms supply some developmental substrate, and impairment phenotypes show cascades across several of these systems. The coupling still doesn't make capacity one category, because the projection targets remain non-subsuming and the stabilizers don't perturb as a single mechanism set. Claims travelling between them need explicit warrant. The Casten / Pinker case from §1 is exactly such a traversal: Casten et al. measure constituent categories 3 and 4; Pinker and Bloom's faculty argument lives at constituent category 1; the evidence in hand doesn't entitle the inference. The Hauser–Chomsky–Fitch / Pinker–Jackendoff exchange is a similar dance about which constituent category the FLN claim actually targets. Section 9 works through the case in detail.

4 IDIOLECT AND ⟨I-language⟩

The grouping around *idiolect* (and Chomsky's *I-language*) covers *language* claims at the individual scale: what a speaker knows, commands, or has internalized. Applied here, the §2 test fails for single-category status. The constituent categories discussed (Chomskyan I-language, Devittian idiolect, and empirical individual variation) share a label and a research community but disagree at the ontological level about what the individual-level entity is, and the disagreement runs through to stabilizers: genetic-cognitive (UG plus maturation) for I-language, convention-based (Lewis-style mutual expectation plus entrenchment) for Devittian idiolect, exposure history and network density for empirical variation. Projection targets diverge correspondingly. Perturbing individual exposure doesn't predict patterned decay across the architectural or convention categories. The grouping is best treated as a neighbourhood at the individual scale.

One constituent category is *I-language* in Chomsky's sense (generative linguistics, philosophy of mind): the mental system in a competent speaker, the steady state reached at the end of acquisition, and the universal grammar shared across the species that constrains which I-languages are humanly possible. Stabilizers proposed for I-language run from genetic endowment to maturational mechanisms to consolidation in adulthood, with linguistic input from the community as the trigger. The minimalist programme's Merge sits here as a candidate computational primitive. An acceptability-judgement asymmetry, acquisition-order pattern, or aphasia profile projects to its replication in related languages, new cohorts, or new patients via UG-constraint ordering; defeated if the pattern is derivable from non-mental mechanisms or alternative frameworks constrain it as tightly.

A second constituent category is the idiolect as Devitt (2008) characterizes it (philosophy of linguistics): the individual's command over a non-psychological realm of physical linguistic entities – sounds, inscriptions, gestures – whose linguistic properties (syntactic, semantic, phonological) are

conventionally established. Devitt's argument turns on three distinctions: between a theory of competence and a theory of its outputs, between structure rules governing the outputs and processing rules governing the exercise, and between a competence respecting the structure rules and one including those rules among its processing rules. The grammar, on this view, is a theory of the linguistic reality, and what it tells us about the speaker's mind is just that the speaker has "something-we-know-not-what" that respects the rules (Devitt, 2008, p. 219). Stabilizers are convention (Devitt invokes Lewis 1969 explicitly), entrenchment of conventional uses, and mutual understanding within speech communities. A structural property of an expression class projects to similar properties in unobserved expressions through Lewis-style convention extension; defeated if the convention has fragmented or the population is too thinly connected for common knowledge.

A third constituent category is empirical individual variation (psycholinguistics, sociolinguistics): differential competence across individuals, in Hymes's heterogeneous-speech-community sense (Hymes, 1972). Bloomfield's contrast between White Thunder, whose Menominee was thin and shaped by acculturation, and Red Cloud Woman, who commanded Menominee, Ojibwa, and Potawatomi fluently, sits here. Dąbrowska's documentation of substantial individual variation in grammatical knowledge (in rule extraction, morphological productivity, and comprehension of complex constructions, with education and written-language exposure as recurring predictors) sits here too (Dąbrowska, 2012). Stabilizers are acquisition history, exposure-driven entrenchment, and individual network density. A speaker's variation pattern projects to similar patterns in similar individuals under conditioning on input history; defeated if the variation is uniformly random or fully explained by population-level convention.

The Devitt–Chomsky exchange is constituent-category distinction work being negotiated in real time. Chomsky takes the individual-level entity to be a mental system; Devitt takes it to be the individual's command over public-language expressions; Collins (2007, 2008) press Devitt on whether there's a non-psychological linguistic reality to study at all; Rey's antirealist argument denies that the relevant linguistic entities exist in either form (Rey, 2006a, 2006b). The HPC discipline doesn't adjudicate the ontological dispute, but it locates where each disputant's claims project: UG-shaped predictions about cross-linguistic stability and maturational trajectory for I-language; convention-extension predictions about novel expressions for the Devittian category; variation patterns under input perturbation for empirical individual variation. Treating data licensed by one as direct evidence for another is the within-neighbourhood slippage diagnosed in §9.

5 COMMUNICATIVE PRACTICE

The grouping of *language* claims about *what speakers are doing when they communicate* carries its own analytical engine, drawn from speech-act theory, conversation analysis, interactional sociolinguistics, and linguistic anthropology, and the engine handles phenomena (illocutionary force, interactional uptake, situated competence, register formation, metapragmatic typification) that no other element of the map captures at the right grain. Applied here, the §2 test fails for single-category status; the constituent categories vary in stabilizers (constitutive rules for speech acts; alignment and turn-taking for interactional organization; conditioned appropriateness for communicative competence; reflexive enregisterment for metapragmatic typification), and their projection targets don't subsume one another. Perturbing one doesn't predict patterned decay across the others: enregisterment can fragment without felicity conditions shifting, and felicity conditions for declarations can shift (legal

performative reform) without conversation-analytic turn-taking machinery breaking. The grouping shares a research community and an analytical engine, but the homeostasis is plural. Treating it as a neighbourhood at the situated-and-conventional scale makes within-grouping traversals legible as moves needing warrant.

One constituent category is speech acts (pragmatics, philosophy of language). Austin (1962) and Searle (1969) showed that uttering can be doing (promising, requesting, asserting, declaring, apologizing) and that the doing is conventional rather than expressive. The illocutionary force of an utterance is constituted by rules at the population level, with felicity conditions specifying what counts as a successful instance. Stabilizers are conventionalized constitutive rules, acquisition of speech-act conventions in the same developmental matrix as grammar, and institutional infrastructure for high-stakes speech-act types (legal performatives, religious oaths, ceremonial declarations). A form deployed by a speaker oriented to community *C*'s speech-act conventions projects to predictable uptake, repair, and performance-as in unobserved exchanges within *C*; defeated if the form fails to count as the speech-act type for participants (mis-coordination, cultural difference, ironic or mock deployment).

A second constituent category is interactional and situational organization (conversation analysis, interactional sociolinguistics). Goffman (1981) on footing and participation frameworks, conversation-analytic work on turn-taking and repair (Sacks & Schegloff, 1979; Schegloff et al., 1977), and interactive-alignment research on dyadic coordination (Pickering & Garrod, 2004) together specify how interaction is sequentially managed: who speaks when, how repair is initiated and completed, how participation roles (speaker, addressee, overhearer, ratified third party) are shifted by footing, and how utterances are understood as moves in a sequentially-organized exchange. *Uptake* is the load-bearing notion: what an utterance does in the interaction depends on how subsequent turns treat it. Stabilizers are interactive alignment at the dyadic level, cultural transmission of interactional conventions, and institutional gatekeeping in high-stakes contexts (courtroom, classroom, medical encounter). An interactional move or sequence projects to predictable response, repair, and participation-framework consequences in similar contexts; defeated if cross-cultural transfer fails systematically or some unmodelled stabilizer is operative.

A third constituent category is communicative competence in Hymes (1972)'s sense (sociolinguistics, anthropological linguistics): the integrated capacity to deploy language as formally possible, feasible, appropriate, and in fact done. Hymes shows that the Chomskyan competence/performance dichotomy is too narrow; what speakers know includes when to speak, when not, what to talk about with whom, when, where, and in what manner. Differential competence within heterogeneous speech communities is the rule, not the exception. Stabilizers are acquisition (children acquire appropriateness rules in the same developmental matrix as grammar), interactive alignment, community transmission of register and address-form conventions, and institutional gatekeeping for some sectors. A speaker's performance across multiple contexts projects to performance in similar contexts and judgements about inappropriate uses, with register, addressee, and discourse community as the conditioning variables; defeated if the conditioning-gain test (§6) returns null or appropriateness rules don't actually condition speaker performance.

A fourth constituent category is metapragmatic typification, the reflexive level at which speakers and communities recognize forms as indexical of speaker attributes. The apparatus is from linguistic anthropology and the third-wave sociolinguistic tradition. Agha (2005) characterizes ENREGISTERMENT as the process whereby distinct forms of speech come to be socially recognized as indexical of

speaker attributes through population-level uptake, producing recognizable enregistered voices. Silverstein (2003)'s indexical orders give the layered structure: an n -th order indexical (a form indexing a population, register, or stance) is always available for re-analysis at $n + 1$, with ideological and meta-pragmatic work driving the re-analysis. Eckert (2008)'s indexical fields capture the constellation of meanings any one form can carry. Stabilizers are reflexive social cognition, looping feedback through speakers' awareness of indexical links, ideological reproduction, and the back-and-forth of convergence and divergence in situated stylistic deployment. A form's indexical field projects to matched-guise inferences in similar community members and to predictable ideological-shift effects; defeated if the field doesn't replicate cross-community or ideology shifts don't track field shifts.

Individuating the constituent categories makes within-grouping traversals visible as moves needing warrant: a speech-act felicity finding doesn't transfer to a register-uptake claim without the meta-pragmatic step; a conversation-analytic repair pattern doesn't transfer to an appropriateness-rule generalization without the four-sector competence frame; a register-recognition study doesn't transfer to a speech-act-felicity claim without the linking inference that the register conditions felicity. The corpus, the treebank, the conversation-analytic transcript, and the LLM output are all analyst's diagnostics. Treating any of them *as* communicative practice substitutes the diagnostic for the maintenance mechanism (§9 catalogues such cases alongside Casten / Pinker).

The framework is compatible with languaging-focused approaches that treat language as situated coordinative activity (Pennycook, 2010): where those approaches take that activity as primary, the constituent categories here specify the stabilizers and projection targets at which it is sustained.

6 POPULATION-LEVEL CONVENTION

Language as population convention sits at the population grain, between the individual idiolect and the institutionally-named variety. It is what Chomsky calls (and dismisses as artificial) *E-language*, what Lewis (1969) theorizes as convention, and what Labov's variationist tradition empirically studies. Santana (2016) rebuts Chomsky's anti-E-language arguments at length, and the rebuttal is largely accepted in this paper. The recurrent constituent categories foregrounded here pursue different research questions and project to different inferences.

One constituent category is convention in Lewis's sense (philosophy of language, formal pragmatics): a regularity such that almost everyone conforms, almost everyone expects everyone else to conform, almost everyone prefers any one to conform conditional on the others, and at least one alternative would meet the same conditions. Linguistic conventions on this analysis are coordination equilibria: speakers conform (use *chien* for dogs in French, *dog* in English, 犬 in Japanese) because they expect others to and prefer to conditional on others doing so. Stabilizers are mutual expectation, communicative success as payoff, and iterated interaction producing common knowledge. A regularity at the population level projects to its persistence under perturbation and to its tracking of the common-interest structure; defeated if the regularity persists despite deviation (so isn't convention-sustained) or the population is too thinly connected for common knowledge.

A second constituent category is empirical speech-community variation (variationist sociolinguistics). Labov's foundational work (on Martha's Vineyard's centralized /ay/, on post-vocalic /r/ in New York City department stores, on the orderly heterogeneity that variable rules describe rather than smoothing away) establishes that variation at the population level is systematic, structured, and conditioned by demographic variables (class, gender, age, region) (Labov, 1972, 2001). The method-

ological backbone is sociolinguistic interviewing combined with multivariate analysis; Tagliamonte and D'Arcy (2007) and others have extended the apparatus to apparent-time and real-time studies of grammaticalization and change. Milroy (1987) adds network density: tightly connected local networks maintain non-prestige variants and resist change-from-outside, while loose networks accept it. Stabilizers are interactive alignment within the community, identity-marking pressures, network density, and iterated transmission. A variable's distribution in interview data projects to similar distribution in similar speakers and to predictable change trajectories under demographic and network perturbation; defeated if the distribution fails to replicate or conditioning doesn't recover the pattern.

A third constituent category is style (the third wave of variation studies, in Eckert's periodization): variables function as stylistic resources carrying ideologically structured social meanings. Eckert (2008) argues that variables don't have fixed meanings but inhabit indexical fields: constellations of ideologically related potential meanings, any of which can be activated in situated use. Building on Silverstein (2003)'s indexical orders, the framework treats stylistic moves as ideological work, with speakers positioning themselves with respect to elements of character, persona, and stance that the variable has been enregistered to index. Eckert's (2008) reanalysis of Labov's Martha's Vineyard fishermen makes the move concrete: the centralized /ay/ wasn't simply claiming Vineyarder identity, it was making a claim about what a Vineyarder is, against the encroachment of mainland tourism. Stabilizers are reflexive metapragmatic awareness, the back-and-forth of convergence and divergence, ideological reproduction, and recurrent stylistic deployment. A variable's indexical field projects to its social meaning in similar contexts and to predictable shifts under ideological pressure; defeated if the field doesn't replicate cross-community or ideological-shift predictions fail.

Existential agreement gives the category its clearest worked HPC audit. Santana's three criteria settle the existence question: English subject-verb agreement exists, is a primary object of study across formal syntax, sociolinguistics, and corpus linguistics, and is reasonably called part of *English*. HPC picks up where Santana leaves off, asking what's maintained at the population level.

PROPERTY COVARIANCE shows up in two patterns: *there's* occurs disproportionately with following plural noun phrases in informal contexts (*there's two men*), and categorical agreement morphology covaries with edited-prose context.

STABILIZERS at the population level fall into two distinct mechanism sets: (a) interactive alignment in casual conversation lets *there's* stabilize as an invariant presentative particle decoupled from the number of the following noun phrase, and (b) institutional editorial gatekeeping in edited prose maintains categorical agreement.

PERTURBATION SENSITIVITY runs through both stabilizers. Weakening either predicts patterned decay: decreasing editorial enforcement (online publishing without copy-editing, for instance) predicts increased *there's two men* in supposedly-edited contexts, and loss of register distinction predicts collapse of the two-system pattern toward whichever variant is more common in a given community.

The PROJECTION target is conditioned, not pooled: a maintenance claim about *there's* licenses predictions about which registers will tolerate it and which won't, and about which speakers will produce or accept it under which conditioning.

The DEFEATER would be cross-corpus failure of the conditioning-gain test: if rates of *there's two men* versus *there are two men* proved unstructured-noise across registers under properly-controlled comparison, the two-register account would lose its grip. Krejci and Hilton (2017) document the structured pattern (with a third levelled *there is* variant alongside the two main ones); Coppock (2018)

shows the parallel double-copula case (*the thing is is that...*) is similarly predictable under spontaneous-planning conditioning.

Santana's three-criteria framework leaves the maintenance question open. HPC supplies the CONDITIONING-GAIN TEST that disciplines what counts as a population-maintained pattern: conditioning on a putative variable should improve prediction over the pooled baseline, and no gain means no mechanism is sustaining the apparent pattern. What's maintained here is two register-conditioned systems, not one pooled system. Naming what's maintained is the section's payoff, and the slippage diagnosed in §9 is what happens when pooled data is read as evidence about a single underlying grammar.

7 NAMED SOCIOHISTORICAL OBJECTS

Language as a named variety (English, French, Quebec French, Oji-Cree, K'iche', Japanese, Cantonese, Modern Standard Arabic, Shilluk) sits at the institutional grain rather than the conventional one. What separates it from §6 is the additional infrastructure: schooling, literacy, codification, legal recognition, language tests, language ideology. The recurrent constituent categories foregrounded here pursue different research questions about the same named objects.

One constituent category is the named object as institutional artefact (sociolinguistics, language policy). Haugen (1966) catalogued the standardization moves that build named languages: selection of a norm, codification of form, elaboration of function, acceptance by the population. Institutional configurations vary widely across the category: state-backed codification, engineered national-language projects, Indigenous-language community standardization paired with language-rights frameworks, and named varieties whose institutional scaffolding is much lighter. National academies, codifying dictionaries, school curricula, language tests, and linguistic-rights legislation are the visible scaffolding where they exist. Stabilizers run from publishing standards through schooling and literacy education to legal and administrative recognition; the intensity varies by variety. A feature institutionally codified as part of named variety V projects to its reliable appearance in V's active institutional channels and to conformity by speakers oriented to V's gatekeepers; defeated if institutional reinforcement is too thin to maintain the predicted patterns or the feature's distribution doesn't track the channels available.

A second constituent category is the named object as identity-marker (linguistic anthropology, sociolinguistics). Named varieties can become emblems of regional, Indigenous, religious, national, or revival identities. The metapragmatic machinery here is enregisterment in Agha's sense: distinct forms of speech come to be socially recognized as indexical of speaker attributes through processes of population-level uptake (Agha, 2005). Once enregistered, the variety can become emblematic of a recognizable persona, position, or stance. Irvine and Gal (2000)'s analysis of iconization (a feature is taken to embody a quality of its speakers) and erasure (heterogeneity within the named group is downplayed in the ideological summary) names the ideological machinery that hardens a variable distribution into an identity-emblem. Hacking (1999) captures the feedback: once speakers learn that a form indexes a named variety, their use shifts in response to that knowledge, and the shift reinforces the index. Stabilizers are reflexive metapragmatic awareness, the back-and-forth of convergence and divergence, ideological reproduction, and looping feedback through speakers' awareness of the named-variety identity. An indexical association of a named variety with an identity projects to identity-claimers' orientation, community recognition, and ideological reshaping of the field; de-

feated if the claim fails to replicate or the named variety isn't doing identity-work for the population claimed.

A third constituent category is the named object as historical kind in Millikan's sense (historical linguistics): a sociohistorical lineage tracing through identifiable earlier states. Old English to Middle English to Modern English; Latin to the Romance languages; Common Slavic to its descendants. The methodology is historical-linguistic reconstruction backed by manuscript and textual evidence. Stabilizers are iterated transmission across generations, standardization decisions that anchor snapshots of the named object (literary canons, dictionaries, school texts), and educational continuity that frames the lineage narrative for each cohort. A feature in a historical state of the named object projects to predictable diachronic changes, related-language correspondences, and reconstructions of earlier states; defeated if the predictions fail against the textual record or the lineage turns out to be substantially constructed by post-hoc narrative.

Count-noun deployment of *language* cuts across these constituent categories. *Cantonese is a language, Welsh is a language, Jamaican Creole is a language*: the count-noun grammar treats the named object as enumerable, instantiable, and bounded. That count-noun availability is itself a stabilizer at the metapragmatic-ideological grain. It underwrites the bureaucratic infrastructure (census categories, language-test certifications, language-rights legislation) and the educational practices (selecting one norm to teach as *the* standard, distinguishing one named language from a related one in school curricula) that the more visible stabilizers operate through. Haugen's standardization moves all presuppose count-noun discreteness; Irvine and Gal's erasure operates on the assumption that one named language is being demarcated from another. The count-noun ideology functions in this category the way enregisterment functions in §5: a reflexive cultural framework that sustains, rather than transparently describes, the named-object pattern. Treating it as one of the category's stabilizers (rather than as pre-institutional reality) keeps the realist commitment located at the right grain and prepares the slippage diagnosis in §9, which traces what happens when count-noun-licensed recognition data is read as evidence about other elements of the map.

The category's reality has been seriously contested. Makoni and Pennycook (2006) frame "disinventing languages" as the recovery of local practice from the named-object overlay; Pennycook (2010) develops the local-practice side of that critique. The critique is partly right: named-language objects aren't natural kinds awaiting discovery, and the institutional apparatus naming, codifying, teaching, and legally recognizing them is doing constitutive work. The HPC reframing locates reality at the right grain: a named object is real as an institutionally-stabilized attractor basin sustained by schooling, literacy, gatekeeping, and identity-work; it isn't real as a pre-existing natural kind, and treating it as one is one of the slippages diagnosed in §9. Reading evidence about it as evidence about capacity, idiolect, or communicative practice traverses boundaries that the institutional stabilizers don't underwrite.

Boyd's HPC framework was built for biological species, where homeostatic mechanisms are causal and the cluster tracks co-occurrences in a mind-independent world. The named-sociohistorical category looks circular by comparison: count-noun ideology is one of its stabilizers, but the ideology depends on the named object being treated as a count noun. Hacking's "looping" captures the structure (Hacking, 1999): speakers' awareness of category membership shapes their use, and the shifted use reinforces the category. This is a difference from biological homeostasis, not a defect: social kinds are partly constituted by the categorization mechanisms that maintain them. Projectibility for so-

cial HPC kinds runs partly through that reflexive categorization, which makes it more historically contingent and field-relative than projectibility for biological kinds. That isn't a deficit: Boyd's field-relative projectibility already lets projection track a field's purposes rather than a mind-independent essence, and Khalidi (2013)'s analysis of social kinds as nodes in causal networks extends the move to kinds whose mechanisms include categorization. The category earns realist standing because the mechanisms (institutional channels, looping feedback, count-noun ideology) sustain the cluster under perturbation, with confidence calibrated to case-specific historical evidence rather than assumed in advance.

8 ABSTRACT FORMAL SYSTEMS

The formal-systems element treats *language* as a mathematical or formal object, and it sits at a different grain from the others. Its working desideratum is extensional adequacy at the formal grain: a grammar or formal model is adequate insofar as it generates the right strings or assigns the right meanings, regardless of whether the same operations are performed in any speaker's head or community. A grammar can generate strings all day without thereby being anyone's language. Phrase-structure rules, dependency grammars, and treebank annotation schemes sit here when treated as formal or engineering representations, not as descriptions of the maintained linguistic patterns picked out by other categories on the map. Its stabilizers are mathematical-disciplinary (definitional stipulation, mathematical inheritance, and the conventions of the relevant mathematical or computational community), not population-level usage. Santana (2016) develops this as the abstract ontology, with Platonism (Katz), formalism (a structuralist-style reading), and fictionalism as candidate philosophical positions on what the formal objects are.

One constituent category is languages as sets of strings (formal-language theory, in the sense cited by Hauser et al. 2002): grammars as finite devices that generate or recognize those sets, and the Chomsky hierarchy that ranks grammars by descriptive power. Natural languages are at least mildly context-sensitive: Pinker and Jackendoff cite the Swiss-German cross-serial dependencies and Bambara reduplication results that put natural languages above context-free in the standard hierarchy. Jäger and Rogers (2012) review the refinements relevant to cognitive science: mildly context-sensitive grammars (Tree-Adjoining Grammar, Combinatory Categorical Grammar, minimalist grammars) above context-free, and the sub-regular hierarchy below regular for local-pattern phenomena. Stabilizers are mathematical inheritance and the disciplinary conventions of formal-language theory. A phenomenon like centre embedding or cross-serial dependencies projects to a hierarchy-level lower bound on the language and on the cognitive mechanism recognizing it; defeated if the formal-language-theoretic characterization is inadequate to phenomena that exceed it or doesn't correspond to the cognitive mechanism in question.

A second constituent category is meanings as model-theoretic objects (formal semantics in the Montague tradition): functions, sets, possible-world structures composed by rules that map syntactic structure to semantic value. The compositionality move treats the meaning of a complex expression as determined by the meanings of its parts and the way they're combined; type-theoretic and intensional developments extend the apparatus to quantification, modality, and dynamic phenomena. Stabilizers are model-theoretic and type-theoretic inheritance. A semantic property of expressions in test cases projects to the same property in cases of similar structure through compositional warrant; defeated if the characterization fails on meaning judgements, universals don't replicate, or phenomena resist

compositional analysis.

A third constituent category is linguistic objects as Platonic abstracta (Katz’s Platonist linguistics, Katz 1981): abstract entities analogous to numbers in mathematical Platonism. The grammar describes those abstract objects; extensional adequacy is achievable without commitment to psychological reality. Stabilizers are the Platonist tradition and disciplinary practice in the (smaller) Platonist-leaning subset of formal linguistics. A structural pattern projects to the same pattern in unattested cases with mathematical-style necessity and counterfactual stability under variation in psychological and sociological inputs; defeated if alternative ontologies (psychological, social, conventionalist, fictionalist) explain the patterns equally well, leaving Platonism no extra explanatory work.

Whether the abstract-formal category’s stabilizers are Platonist or constructivist is a separate question from whether the category projects. The HPC discipline is neutral. Mathematical inheritance and definitional stipulation work whether mathematical objects are mind-independent abstracta (Platonism) or mind-dependent disciplinary constructions (constructivism); both views recognize the same theorems and the same disciplinary continuity, even when they disagree about the metaphysics. The category passes the projectibility check on either reading, because what the check tests is whether learning about one formal-language-theoretic structure supports inferences about others, which is a fact about the practice rather than its philosophical foundation.

A fourth constituent category is language as computational artefact (NLP working usage, Carnap-style methodological idealization at the engineering scale). Parsers, generators, machine-translation systems, and (now) large language models all treat formal models of language as the working object, with success measured by performance benchmarks rather than psychological or sociological fidelity. Stabilizers are engineering tradition, computational-linguistic disciplinary practice, and the cumulative apparatus of mathematical inheritance from formal-language theory and formal semantics. A system’s performance on a class of tasks projects to similar performance and predictable transfer or failure outside the class; defeated if transfer fails in ways the model doesn’t predict or the success comes from properties unrelated to the formal characterization.

The category’s borderline verdict in Table 1 reflects an asymmetry across these constituent categories. Formal-language theory and formal semantics propagate forward to NLP engineering practice: a context-sensitivity result constrains what NLP architectures must handle, a compositionality result constrains what semantic representations must encode. NLP engineering practice doesn’t reliably propagate back: a benchmark improvement, or a transformer’s behaviour on a class of tasks, doesn’t constrain formal-language-theoretic claims about the underlying object. The asymmetry preserves single-category status – the shared mathematical-disciplinary backbone and overlapping projection target license inferences across the constituent categories – at lower confidence than the symmetric-coupling categories pass with. Claims originating in NLP-engineering practice need extra warrant before being read as evidence about formal-grammar status; §9 catalogues this NLP-into-formal-grammar move alongside the broader formal-system-into-empirical-object slippage.

The category’s typical traversal is to capacity. Hauser et al.’s Hypothesis 3, that the faculty of language in the narrow sense (FLN) consists of recursion plus the interfaces with sensory-motor and conceptual-intentional systems, is a worked example. Recursion is well-defined as a formal-language-theoretic property. FLN, in Hauser et al.’s sense, is a cognitive-architectural posit in capacity’s abstract-computational category. Pinker and Jackendoff argue that the recursion-only thesis transfers a §8 formal-language-theory characterization to a §3 abstract-computational-architecture claim without

doing the work that bridge would need: phonology has discrete infinity but isn't technically recursive; words encode language-specific syntactic information not captured by recursion alone; case, agreement, and most of the syntactic apparatus beyond hierarchical phrase grouping are language-specific. The disagreement is partly empirical, partly methodological. The HPC discipline names the methodological half: a formal-language-theoretic property is the kind of thing that's stabilized by mathematical inheritance, and a faculty-architecture claim is the kind of thing that's stabilized by genetic and developmental machinery; the two stabilizer kinds don't underwrite each other's inferences, and §9 catalogues this category-to-neighbourhood move alongside the Casten / Pinker case.

9 SLIPPAGE

The taxonomy from §§3–8 earns its keep when it does diagnostic work Santana-style pluralism leaves underspecified. “Slippage” is the recurring move where evidence licensed by the stabilizers of one element gets read as evidence about another, whether across the map or within a neighbourhood between constituent categories that don't share a stabilizer set. HPC supplies the discipline: when stabilizers don't transfer across the relevant boundary, the inference doesn't either. This section works two cases in detail, catalogues other slippage classes, and closes with the conditions under which evidence *can* legitimately travel.

9.1 THE CASTEN / PINKER CASE

The Casten / Pinker case from §1 is the worked biocognitive instance. Pinker, citing Casten et al. (2026) as vindication of Pinker and Bloom (1990), reads it as evidence for a uniquely human modular language faculty.

Figure 1 schematizes projection warrant in the case, with the faculty target split into Pinker–Bloom broad and HCF narrow versions (Hauser et al., 2002; Pinker & Bloom, 1990). F1 anchors evidence within the developmental-acquisition category; HAQER-like sequence similarity across nonprimate mammals anchors evidence within the comparative-biology category. Solid links mark locally projectible evidence within each stabilizer profile; dashed links mark unsupported bridges to the faculty targets. The narrow-FLN target is a further inferential step Pinker's tweet doesn't take explicitly but the same evidence is often used to license.

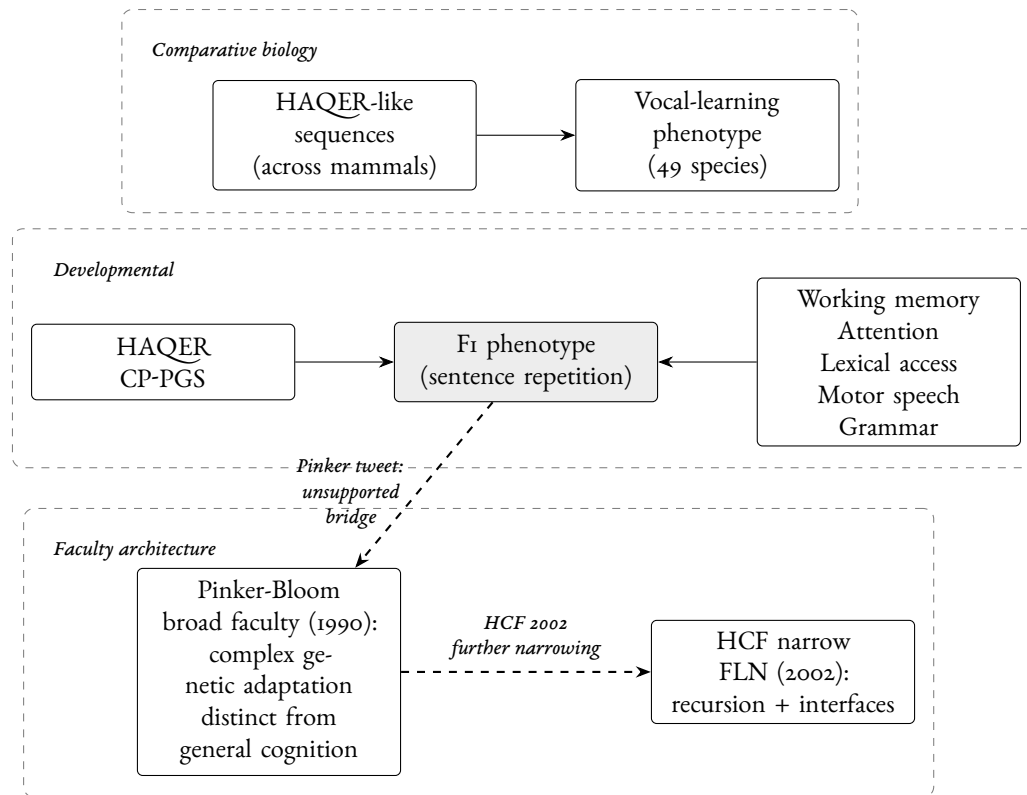


Figure 1: Projection-warrant schematic for the Casten / Pinker slippage, with the faculty target split into Pinker-Bloom broad and HCF narrow versions. Solid links mark locally warranted projection within constituent categories. The dashed link from F1 to the broad faculty marks the inference Pinker’s tweet makes; the dashed link from broad to narrow marks the further narrowing step (Hauser et al., 2002). None of the three bridge-principle conditions is satisfied for either dashed link. The schematic is local to this biocognitive case; sociocultural slippages need feedback structure this kind of diagram doesn’t supply.

The charitable reading doesn’t rescue the inference. Pinker’s 280 characters say “suggest we were right”, not “prove”, and the Pinker–Bloom thesis (1990) covered the broad faculty (a complex genetic adaptation including vocal-learning architecture, sensory-motor systems, and conceptual-intentional interfaces), not just the narrow recursive computation HCF later isolated. A charitable reconstruction reads Pinker as updating on the broad faculty, and Figure 1 accommodates this by splitting the faculty target.

The issue isn’t whether F1 causes the faculty; it’s what an F1 finding licenses an update on. F1’s stabilizers underwrite developmental-acquisition and comparative-biology claims at the constituent-category grain. The regulatory architecture is shared at the cluster level with cetaceans, pinnipeds, and bats; whether human-specific tuning is layered on that backbone is a separable empirical question Casten et al.’s evidence doesn’t bear on. F1 loads on attention, working memory, lexical access, and motor speech alongside grammatical processing. The update Pinker is entitled to is a weak one on the shared-mammalian component of the broad faculty; the broad-faculty-as-complex-human-adaptation the Pinker–Bloom thesis was arguing for would need extra bridging evidence the Casten

paper doesn't supply. The slippage is the move from the smaller update to the larger thesis.

9.2 THE CANTONESE CASE

A second worked case shifts the slippage to sociocultural ground at full audit length. The named sociohistorical object *Cantonese* is recognized in Hong Kong census language/dialect categories, positioned in Hong Kong education policy, and enregistered as indexical of Cantonese-speaker identity in Hong Kong (Census and Statistics Department, Hong Kong Special Administrative Region, 2022; Lee & Leung, 2012). The named object earns category status at the institutional grain (§7's audit). Census recognition data is the route through which the slippage to population convention appears, and the audit walks the same five elements as §6's existential-agreement case.

PROPERTY COVARIANCE within the named-object frame is real at the institutional grain: reported Cantonese use in census categories, school exposure through Hong Kong's language-in-education policy, and identity-marking work co-occur where institutional channels make *Cantonese* a recognized named object.

STABILIZERS come from §7's institutional, identity, and descended-kind strands plus count-noun ideology: institutional ratification (Hong Kong census classification; Hong Kong education-policy positioning; diaspora census channels), identity-marking through enregisterment (Agha, 2005), looping feedback (Hacking, 1999), and the count-noun ideology that licenses bounding and enumeration. They are institutional and metapragmatic, not interactional-coordinative.

PERTURBATION SENSITIVITY runs through institutional-regime shifts. Shift the regime (post-handover Hong Kong language policy, Putonghua-promotion pressures, or census-question design in diaspora contexts) and the named-object category can reconstitute at a different intensity but hold (Lee & Leung, 2012; Statistics Canada, 2022). A putative *Cantonese-as-population-convention* category doesn't reconstitute under the same perturbations, because it wasn't maintained as one coordinated regularity in the first place.

PROJECTION runs to institutional reach (where Cantonese is reported, taught, or institutionally recognized) and to identity-recognition (who claims Cantonese-speaker status, what indexical work the claim does). It doesn't extend to population-convention rates, individual-cognitive-system uniformity, or speech-act felicity conditions in interaction.

The DEFEATER would be a count that holds without institutional ratification, or one that fails to track speakers' own identity-claims. Neither obtains: institutional ratification and identity-claim are what the count is tracking.

The slippage is reading recognition data as evidence about population convention. "X million speakers of Cantonese" gets read as evidence about Cantonese-as-population-convention, as if a single Lewis-style coordinative regularity unites the population. What census instruments measure is institutional recognition filtered through enregistered identity-claims and count-noun ideology. The named-sociohistorical stabilizers generate the recognition data; the population-convention stabilizers (interactive alignment, transmission, network density) would have to generate any convention claim; the two stabilizer sets don't underwrite each other. Pennycook's case against named-language reality lands here: treating count-noun ideology as transparent description of pre-institutional regularity is the move HPC blocks. Counting speakers isn't the same thing as identifying a convention.

None of the three bridge-principle conditions is satisfied for the recognition-to-convention inference. Linked stabilizers: institutional channels don't causally contribute to interactive alignment in any general way; they shape what speakers report doing, not the population-level regularity. Projection-

target subsumption: institutional reach doesn't subsume coordinative regularity. Auxiliary linkage evidence: a bridge would require independent evidence that institutional ratification predicts interactional coordination at the population grain. That evidence is sometimes available (standard-variety schooling can shift convention rates over generations) but has to be supplied case-by-case. Without the bridge, the slippage is the sociocultural counterpart of the Casten / Pinker case: warrant-chain expansion across boundaries, with element-specific stabilizers doing constitutive work the reading erases.

9.3 RECURRENT SLIPPAGE CLASSES

The same pattern recurs across other element pairs. *Idiolect into population convention*: I-language data is presented as evidence about (or against) E-language regularities, or population-level variation is read as direct evidence about individual mental grammars; the stabilizers operate at different grains. *Named object into convention*: a feature institutionally codified as part of a named variety is taken to be a population-level convention, ignoring that institutional channels have different reach than conventional regularities. *Formal system into empirical object*: formal-language-theoretic results get read as claims about the cognitive faculty, smoothing over the gap between mathematical-disciplinary and individual-cognitive stabilizers. The HCF "recursion only" thesis is partly this last slippage: a formal-language-theoretic property in §8 transferred to an architectural claim in §3.

A second slippage type is operationalization slippage: the analyst's diagnostic substituted for the element itself. "The corpus IS the language" reads cross-corpus convergence (a sign of shared annotation conventions) as evidence about population convention. "Competence IS the language" reads idealized acceptability judgements as direct windows onto idiolect. "The treebank IS the grammar" reads parsed structures as claims about the underlying system rather than the annotation scheme. "The LLM output IS the language" reads generation traces as data about communicative practice or capacity, conflating training-and-tuning artefacts with situated interactional work. In every case the analyst's diagnostic is treated as the maintenance mechanism. Corpora, treebanks, judgements, and model outputs are useful instruments. They make poor ontologies.

Grammaticality, taken as a worked locus, exhibits a related family of traversals. Acceptability ratings get read as direct evidence about grammaticality, conflating a feeling-dimension diagnostic with a claim about the licensing dimension. Processing difficulty (centre-embedding, garden paths) gets read as evidence about grammatical status, transferring capacity data to a convention claim. Corpus frequency is treated as automatic evidence of grammatical status, although high-frequency errors fail to entrench and rare-but-licensed forms persist. Formal-grammar generation is taken as evidence about community licensing, and named-language norms ("standard English says X") as evidence about convention. Each crosses an element boundary; together they form a methodological family that the bridge principles below adjudicate. Grammaticality lives primarily in §6's convention category, with bridges to capacity (processing), communicative practice (situated response), named objects (institutional norms), and formal systems (generative grammars).

9.4 BRIDGE PRINCIPLES FOR LEGITIMATE TRAVERSAL

Slippage diagnosis would be hollow without a positive account of when evidence *can* travel across the map. Three conditions earn the warrant. **DEMONSTRABLY LINKED STABILIZERS**: A's maintenance mechanisms causally contribute to B's, with "demonstrably" requiring at least one mechanistic study and warrant scaling to typological breadth, replication, and the directness of the causal

route. Population-level convention shapes individual idiolect through transmission and entrenchment; once that linkage is established, population data informs idiolect inferences with scope set by mechanism reach. PROJECTION-TARGET SUBSUMPTION: A's projection target includes B's, with strict subsumption licensing the strongest inferences and partial overlap licensing weaker ones. Within capacity, the vocal-learning regulatory architecture sometimes underwrites the developmental-acquisition phenotype where regulatory predictions include input-driven trajectory as a special case. AUXILIARY LINKAGE EVIDENCE: independent evidence ties A's stabilizers to B's outputs in a particular case, carrying the inference without the boundary being crossed; auxiliary linkage is case-specific and doesn't generalize. The three principles are schemata, not standalone tests: they require case-specific operationalization, and the Whorfian bridge below works one such operationalization. Bridge principles do for cross-element traversal what Santana's no-isolationism stricture demands: the elements interact, but the interaction has to be argued, not assumed.

Whorfian hypotheses give the bridge apparatus a positive case. Consider a schematic claim, of the sort a relativity study might try to establish, that some population-convention feature (for example a colour category, evidential marking, or gender system) shapes some capacity-cluster category (for example perception, source attribution, or grammatical-attribute recall). If established, that is a demonstrably-linked-stabilizers bridge: population-convention stabilizers (transmission, entrenchment, recurrent use) causally contribute to capacity-constituent stabilizers (developmental input shaping perceptual or memorial categories). The bridge holds when the linkage is empirically demonstrated: the feature has to be actually conventional rather than pan-human; the effect has to replicate under matched controls; and a plausible causal route through experience has to be in evidence. Where the linkage is demonstrated, the framework licenses the bridge; where it isn't, the diagnosis is the Casten / Pinker pattern. The bridge principles aren't only inference-blockers; they license inferences that the underlying mechanism work supports.

The diagnostic and the bridge principles work as a pair. Naming the slippage in HPC terms makes the move legible as a methodological failure rather than a rhetorical glide. Naming the conditions for legitimate traversal protects pluralism from collapsing into isolated parishes. Together they let *language* do the cross-disciplinary work the umbrella label is for, without smuggling unearned inferences across map elements that don't share the relevant stabilizers.

10 THE UMBRELLA LABEL

The umbrella label *language* keeps its place across the language sciences because it does coordinative work no element-label alone can do. Linguists, philosophers of mind, anthropologists, sociolinguists, computer scientists, and educators converge on the term because it gestures at the field of inquiry they all participate in, even when their specific categories and projection targets diverge. The convergence is genuinely useful: cross-disciplinary work depends on shared vocabulary, and replacing *language* with element-labels in every cross-field context would be both impractical and rhetorically self-defeating.

The coordinative role doesn't license a realist commitment. For the umbrella to be a higher-order HPC category, it would need its own projection target (inferences not already licensed by some individual element) and its own higher-level stabilizers. Neither is in evidence. The map's projection targets concern capacity-constituent architectures, idiolect's constituent categories, communicative-practice constituent categories, population-level convention, named-variety institutional infrastructure, and formal-system extensional adequacy. None has the umbrella as its target; the umbrella

names the shifting, partial overlap among them, and the stabilizers are the mechanisms catalogued in §§3–8. Treating the umbrella as a higher-order kind would invite exactly the inflation pluralism has to resist. The categories and neighbourhoods are the load-bearing units. The umbrella does the work of letting them sit on the same conference programme.

II CONCLUSION

The question *What is language?* has been a productive provocation across the language sciences for at least a century. As a question to be answered straightforwardly, it has also been an obstacle: positive answers in one category get presented as the answer that supersedes others, and reception traffic across category boundaries goes unnoticed. Three replacement questions do better.

The first is which element of the map is in play. *Language* as deployed in a given research context picks out one or more elements, or one or more constituent categories within a neighbourhood, or a new element awaiting addition. Naming what's in play specifies the stabilizers under discussion and the inferences they underwrite.

The second is what projection target the classification licenses. A comparative-biology claim licenses inferences about cross-species regulatory architecture. A claim about population-level convention licenses inferences about community coordination patterns under conditioning. A claim about a named sociohistorical object licenses inferences about institutional reach. Conflating projection targets is the engine of the slippages catalogued in §9.

The third is what slippage risks attend the use of the word in a given case. Some elements are adjacent enough that traversal is warranted on the bridge principles of §9; others have stabilizer profiles that make traversal rarely warranted. Naming the risks, and the conditions for warrant, protects pluralism from collapsing into isolated parishes or unearned cross-element inference.

The taxonomy is provisional. New elements are admissible if they pass the projectibility and homeostasis tests; existing elements can be split, merged, or demoted to neighbourhoods, as capacity, idiolect, and communicative practice already have been. What working scientists do when they say *language* should be diagnosable. The map is the diagnostic instrument, and the projectibility-and-homeostasis discipline is its calibration. Name the element. Name the projection target. Name the slippage risk.

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